Creating a Database Instance in the Cloud with Amazon RDS

Manage Relational Databases

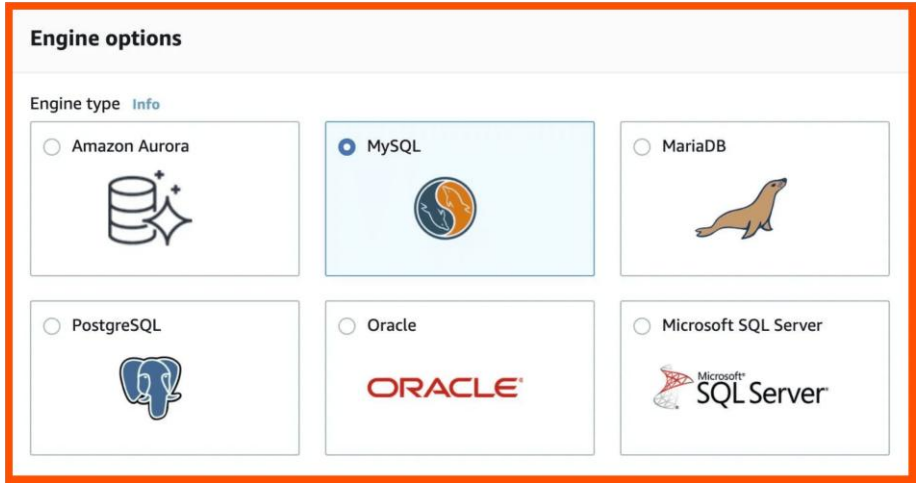


Figure 1. Amazon RDS Management Console

What is Amazon RDS?

- **Definition:** Amazon Relational Database Service (RDS) is a managed service that simplifies setting up, operating, and scaling a relational database in the cloud.
- **Managed Service:** Handles time-consuming tasks like hardware provisioning, database setup, patching, and backups.
- **Scalability:** Easily scale your database's compute and storage resources with a few clicks or API calls.

Key Benefits

- **Multiple Engines:** Supports MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server.
- **High Availability:** Multi-AZ deployments provide enhanced availability and durability.
- **Security:** Provides network isolation, encryption at rest, and encryption in transit.
- **Cost-Effective:** Pay only for the resources you consume with no long-term commitments.

Step 1: Accessing the Amazon RDS Console

Navigating the AWS Management Console

- **Sign In:** Access the AWS Management Console at **aws.amazon.com/console** using your credentials.
- **Search for RDS:** In the search bar at the top of the console, type **"RDS"** .
- **Select Service:** Choose **RDS (Managed Relational Database Service)** from the search results.
- **RDS Dashboard:** You will be redirected to the Amazon RDS dashboard, where you can manage your database instances.

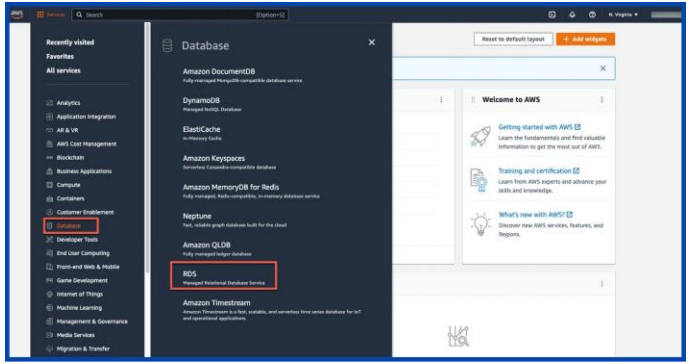


Figure 3. Searching for RDS in the AWS Console

Step 2: Initiating the Database Creation Process

Starting the RDS Configuration Wizard

- **Databases Section:** In the RDS navigation pane on the left, click on **"Databases"** .
- **Create Database:** Locate and click the **"Create database"** button, typically found in the top right or center of the page.
- **Wizard Launch:** This action will launch the database creation wizard, which will guide you through the necessary configuration steps.

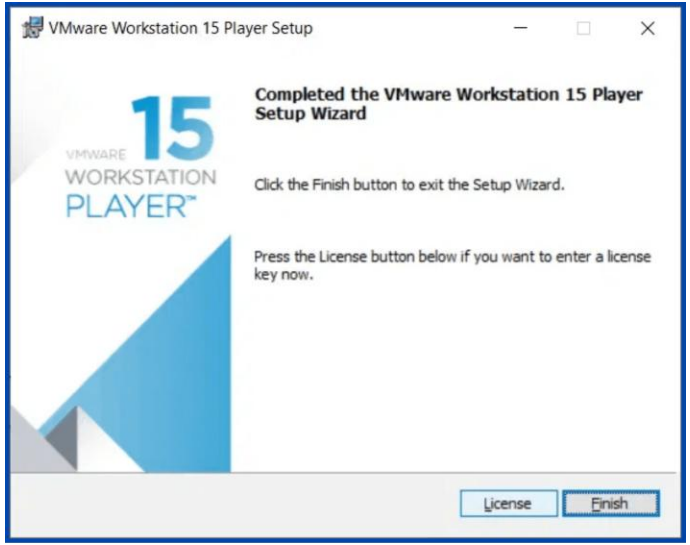


Figure 4. Initiating Database Creation in the RDS Dashboard

Step 3: Selecting the Database Creation Method

Choosing Your Configuration Approach

- **Standard Create:** This method provides full control over all configuration options, including availability, security, and maintenance.
- **Easy Create:** This method uses recommended best-practice configurations for most database settings, simplifying the process.
- **Our Choice:** For this guide, we will use **Standard Create** to explore the various configuration options available in Amazon RDS.

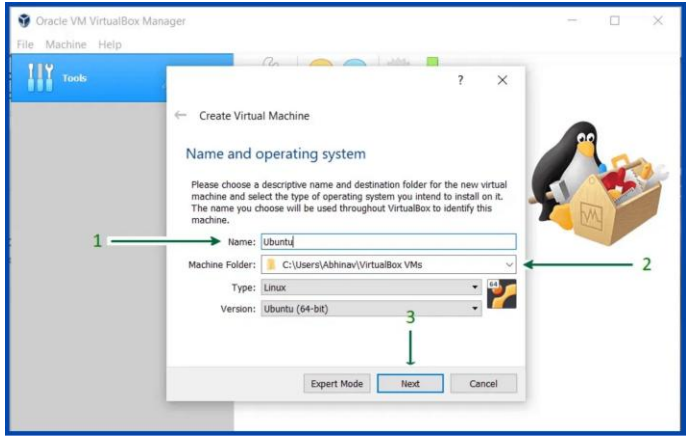


Figure 5. Selecting the Creation Method in the AWS Console

Step 4: Choosing Your Database Engine

Selecting the Right Technology for Your Application

- **Amazon Aurora:** A MySQL and PostgreSQL-compatible relational database built for the cloud.
- **Open Source Engines:** Full support for **MySQL**, **PostgreSQL**, and **MariaDB**.
- **Commercial Engines:** Managed instances for **Oracle** and **Microsoft SQL Server**.
- **Selection Criteria:** Choose based on application compatibility, performance requirements, and licensing needs.

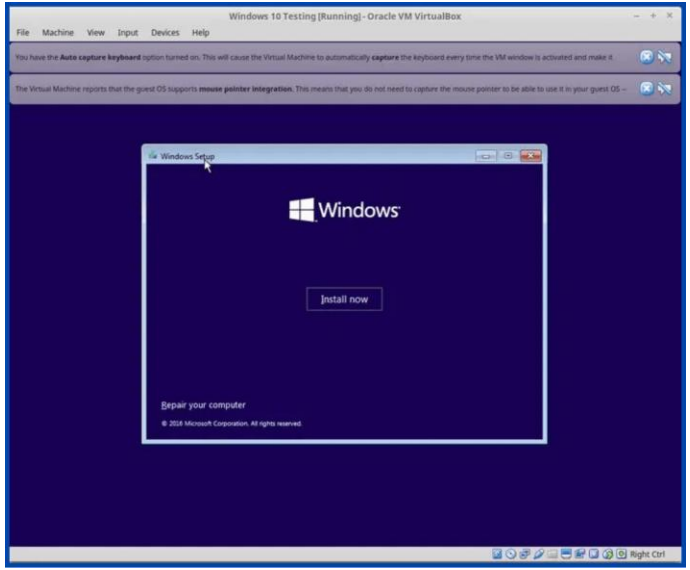


Figure 6. Selecting a Database Engine in the AWS Console

Step 5: Configuring Core Database Settings

Defining Identity and Credentials

- **Templates:** Choose a template that matches your use case: **Production** (high availability), **Dev/Test**, or **Free Tier** (for learning).
- **DB Instance Identifier:** Provide a unique name for your DB instance. This name is used to identify the instance in the AWS console and CLI.
- **Master Username:** Set the primary login name for the database administrator. This user will have full administrative privileges.
- **Master Password:** Create a strong, secure password for the master user. You can also choose to have AWS auto-generate a password for you.
- **Security Note:** Ensure you record these credentials securely, as they are required to connect to your database once it is active.

Step 6: Configuring Instance and Storage

Allocating Compute and Storage Resources

- **DB Instance Class:** Select the compute and memory capacity. Options range from (Free Tier) to high-performance classes. **db.t3.micro**

- **Storage Type:** Choose between **General Purpose SSD (gp2/gp3)**, Provisioned IOPS SSD, or Magnetic storage.

- **Allocated Storage:** Specify the initial size of your database (e.g., 20 GiB). You can enable **Storage Autoscaling** to handle growth.

- **Multi-AZ Deployment:** For production workloads, choose **"Create a standby instance"** to ensure high availability across Availability Zones.

Figure 8. Configuring Instance Class and Storage

Step 7: Setting Up Network and Security

Configuring Connectivity and Access Control

- **Virtual Private Cloud (VPC):** Select the VPC where your database will reside. This defines the virtual network environment.
- **Subnet Group:** Choose a DB subnet group that defines which subnets and IP ranges the database can use.
- **Public Access:** Choose **"No"** to restrict access to within the VPC, or **"Yes"** to allow connections from the public internet.
- **VPC Security Group:** Act as a virtual firewall to control the traffic that is allowed to reach your database instance.

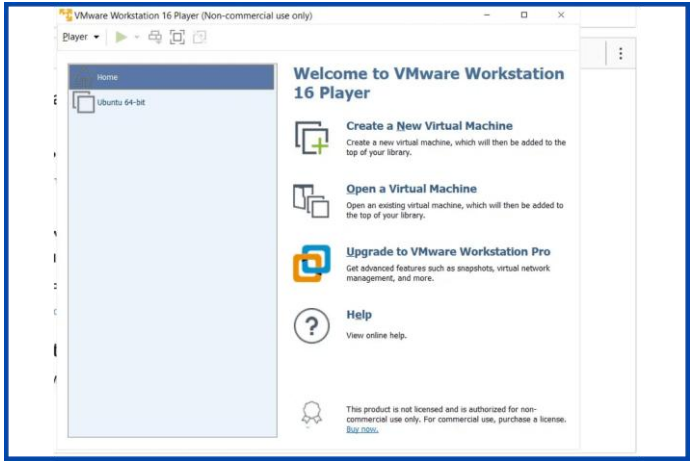
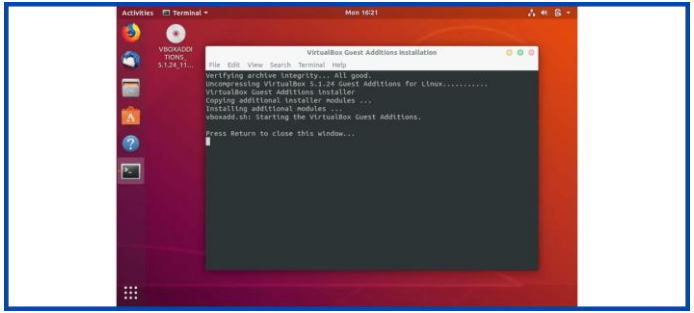


Figure 9. Configuring Connectivity in the AWS Console

Step 8: Finalizing Authentication and Creation

Securing and Launching Your Instance

- **Database Authentication:** Choose between **Password authentication** for more secure access **IAM database authentication** management.
- **Additional Configuration:** Expand this section to verify settings for backups, monitoring, and maintenance windows.
- **Review Costs:** Check the estimated monthly costs displayed at the bottom of the page based on your selections.
- **Create Database:** Click the **Create database** button to start the provisioning process.



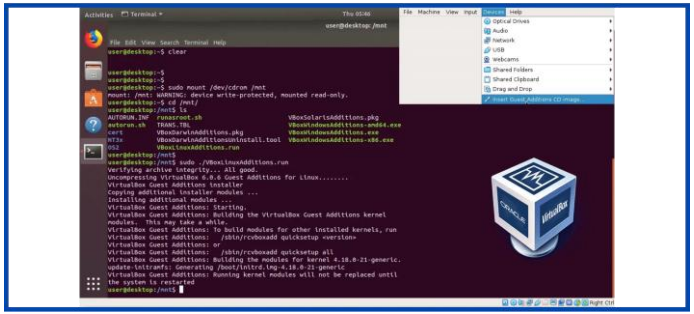


Figure 10. Finalizing Authentication and Creation

Step 9: Viewing and Connecting to Your Database

Accessing Your New Cloud Database

- **Monitor Status:** The database will take a few minutes to initialize. Monitor the **Status** column in the RDS dashboard until it shows **"Available"**.
- **Find the Endpoint:** Click on the database name to view its details. Under the **Connectivity & security** tab, locate the **Endpoint** and **Port**.
- **Database Client:** Use a database management tool (e.g., MySQL Workbench, pgAdmin, or DBeaver) to connect to the database using the endpoint, port, and master credentials.
- **Security Group Check:** Ensure your local IP address is allowed in the **Inbound Rules** of the VPC security group associated with the database.
- **Connection String:** For application development, use the endpoint in your connection string (e.g., **jdbc:mysql://[endpoint]:3306/[db_name]**).

Conclusion: Your Managed Database in the Cloud

Summary of Amazon RDS Benefits and Next Steps

- **Simplified Deployment:** We have successfully navigated the process of creating a robust, cloud-based relational database instance in just a few steps.
- **Operational Excellence:** By using Amazon RDS, you offload the heavy lifting of database administration, including patching, backups, and hardware management.
- **High Availability:** Features like Multi-AZ deployments and automated snapshots ensure your data is secure and always accessible.
- **Next Steps:** Now that your database is active, you can connect your applications using the provided endpoint and begin monitoring performance through the RDS dashboard.
- **Continuous Optimization:** Regularly review your instance metrics to scale resources as your application grows, ensuring optimal performance and cost-efficiency.